

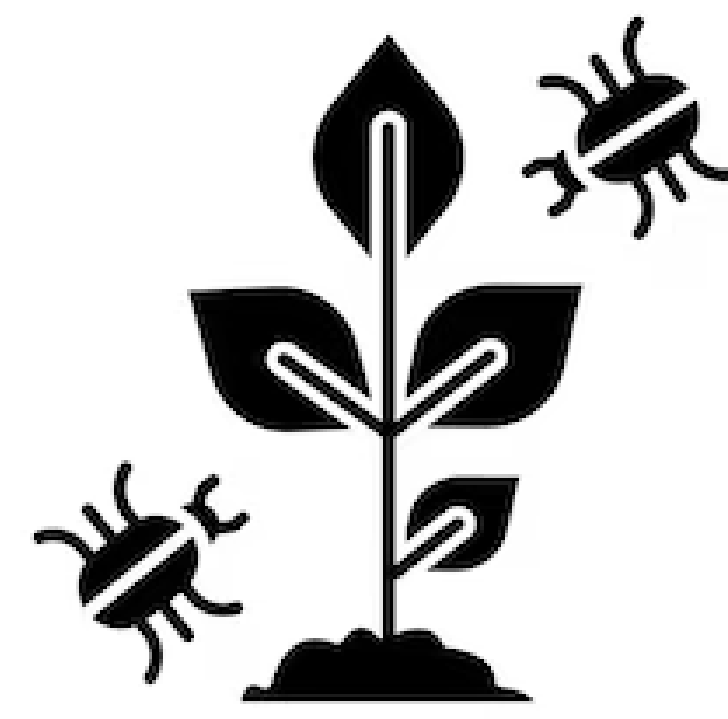


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EVALUATING PRETRAINED ARCHITECTURES FOR AGRICULTURAL PEST CLASSIFICATION ACROSS DIVERSE DATASETS

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Introduction

Pests are a persistent challenge for farmers, gardeners, and other agricultural professionals. Effective pest management aims to minimize pesticide use and cost while maintaining healthy and productive crops. Traditionally, pest identification has relied on human expertise, but advances in technology has allowed assistance through machine and deep learning models.

Although existing pest classification models can achieve high accuracy, many are trained on specific datasets and may struggle to generalize to new environments. They can also be sensitive to real-world conditions such as background noise and image variability, reducing performance.

This project aims to better understand the effectiveness of modern image classification methods for pest identification by comparing two architectures, *Convolutional Neural Networks* (CNNs) and *Vision Transformers* (ViTs), on two new pest image datasets to help guide decision making efforts.

Methodology

We conduct a comparative analysis between two experiments: (1) zero-shot inference and (2) fine-tuning to answer the question:

Which architecture generalizes to realistic, noisy pest datasets better?

Models

1. **CNN:** *ResNet-50* - Available on Pytorch, 50 convolutional layers, pre-trained on ImageNet-1k dataset
2. **ViT:** *ViT-B/16* - Available on Hugging Face(google/vit-base-patch16-224-in21k) and PyTorch, base architecture with 16x16 pixel patches, an input resolution of 224x224, pre-trained on ImageNet-21k dataset

Datasets

1. ***Dangerous Farm Insects Dataset*** (Kaggle)
 - 1,591 relatively clean images
 - 15 classes, $\approx$ 100 images per class
2. ***Pest Dataset V2*** (Kaggle)
 - 34,200 images split into training, validation, and testing data
 - 19 classes, $\approx$ 150 images per class

Data Preprocessing

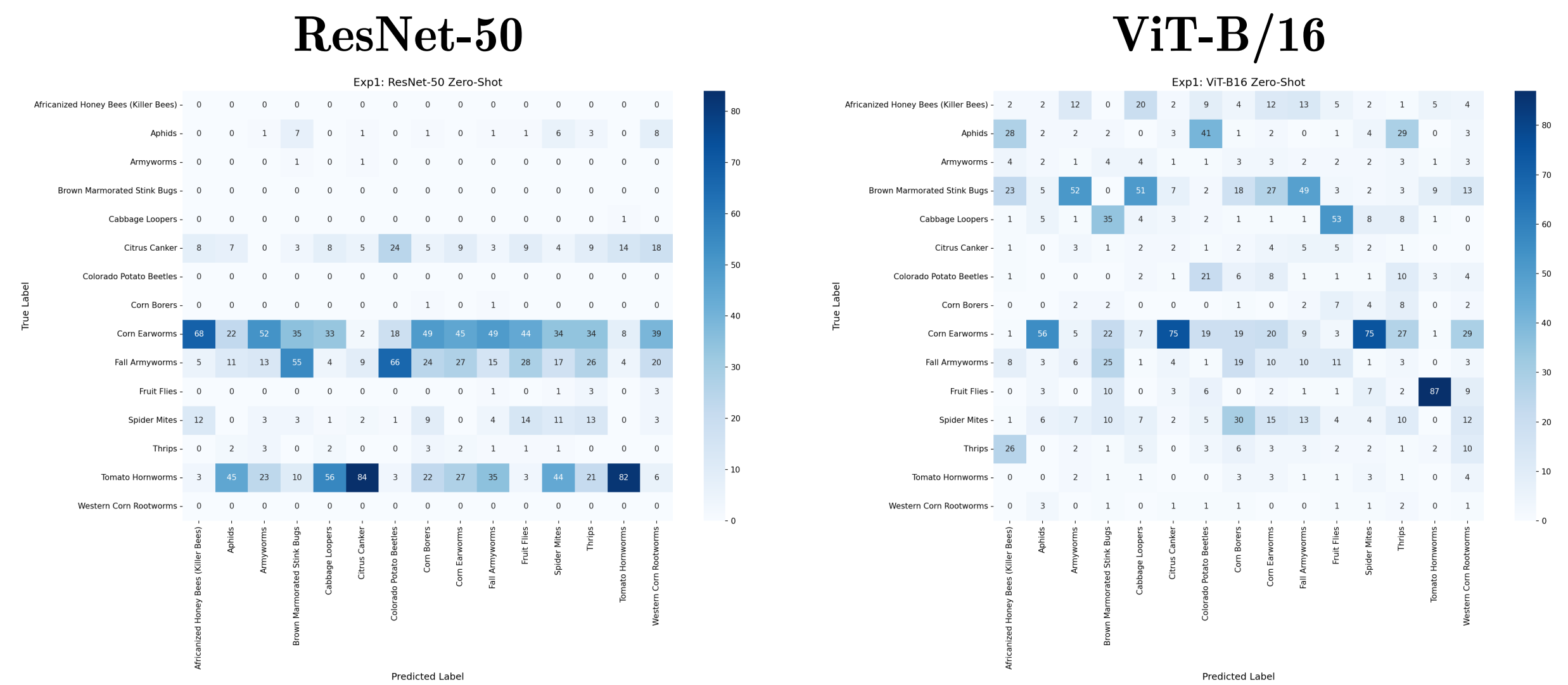
To match the input requirements of ResNet-50 and ViT-B/16 and ensure a reasonable comparison, all images are

- resized such that the shorter size is 256 pixels,
- center-cropped to 224 $\times$ 224,
- converted to tensors,
- normalized using ImageNet mean and standard deviation.

First, no augmentation is applied for without training evaluation.

Then, augmentation is performed to conduct the fine-tuning and transfer learning process of the pest-specific datasets.

Exp 1: Zero-Shot (Dangerous Farm Insects)

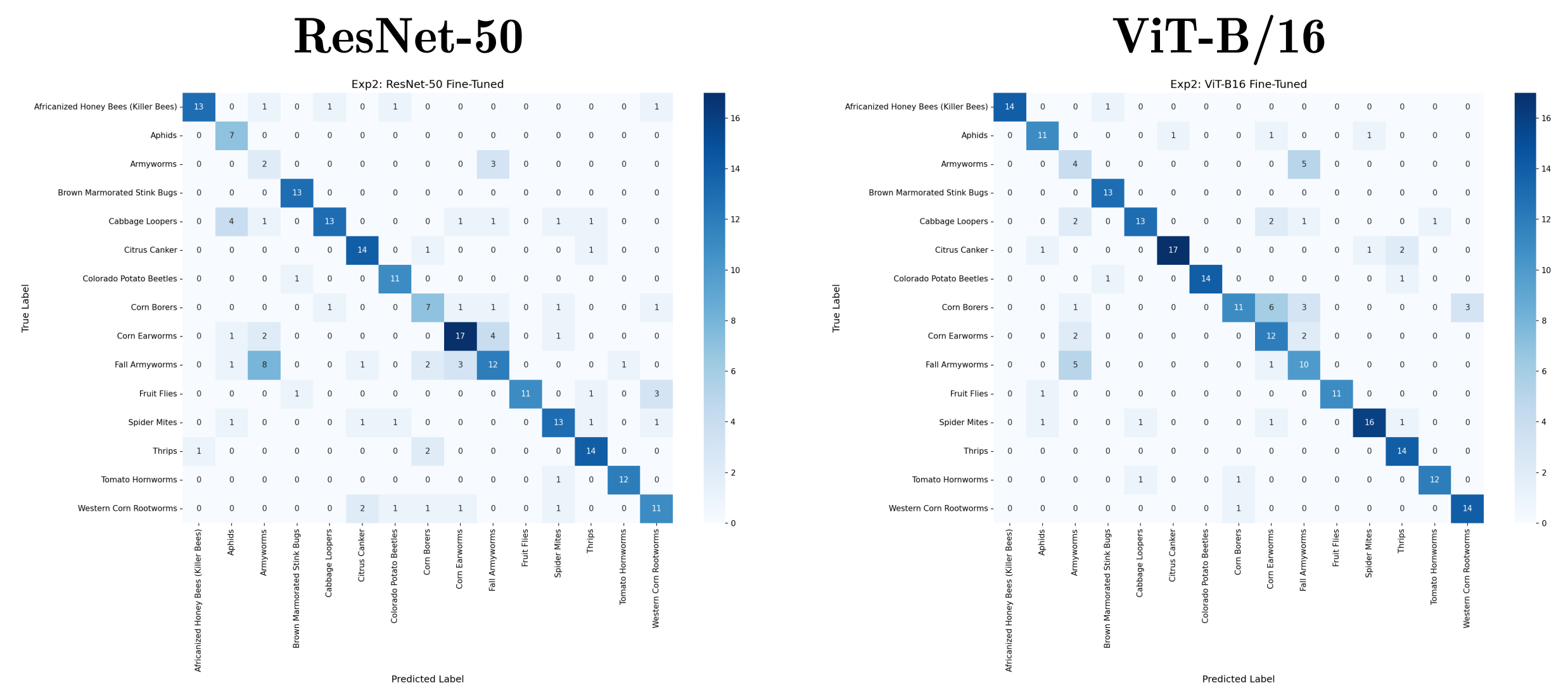


Accuracy: 0.1007
Macro F1 Score: 0.0446

Accuracy: 0.0443
Macro F1 Score: 0.0419

Both the CNN and Vision Transformer did not perform well on the Dangerous Farm Insects dataset. Classification accuracy was slightly higher for ResNet-50 than ViT-B/16, however, it is evident both models at best were guessing from the inconsistency of the confusion matrices. ResNet-50 labeled most of the data into three of the fifteen classes, but predicted them to be spread across all classes. The ViT-B/16 model has more scatter across classes. This is a good baseline, but no meaningful comparison can be made.

Exp 2: Fine-Tuned (Dangerous Farm Insects)

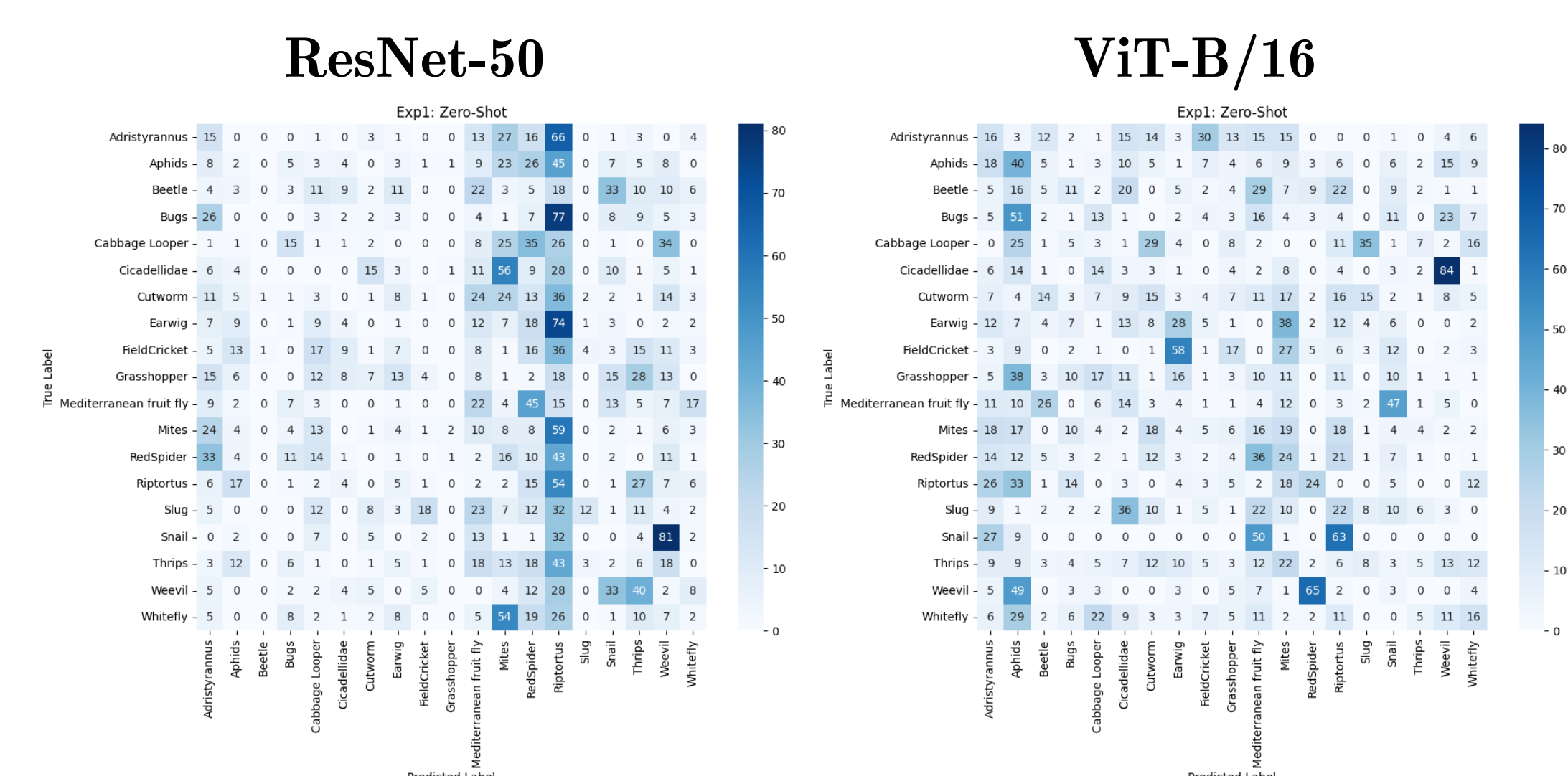


Accuracy: 0.7143
Macro F1 Score: 0.7121
Early Stopping at Epoch 8

Accuracy: 0.7815
Macro F1 Score: 0.7857
Early Stopping at Epoch 10

After fine-tuning, the CNN and Vision Transformer ViT-B/16 model performed well shown by the strong diagonal in both of the confusion matrices. Classification accuracy of ViT-B/16 is stronger likely due to the attention mechanism in the transformer architecture which allows the model to pick up broader details and context that a CNN cannot.

Exp 1: Zero-Shot (Pest Dataset V2)

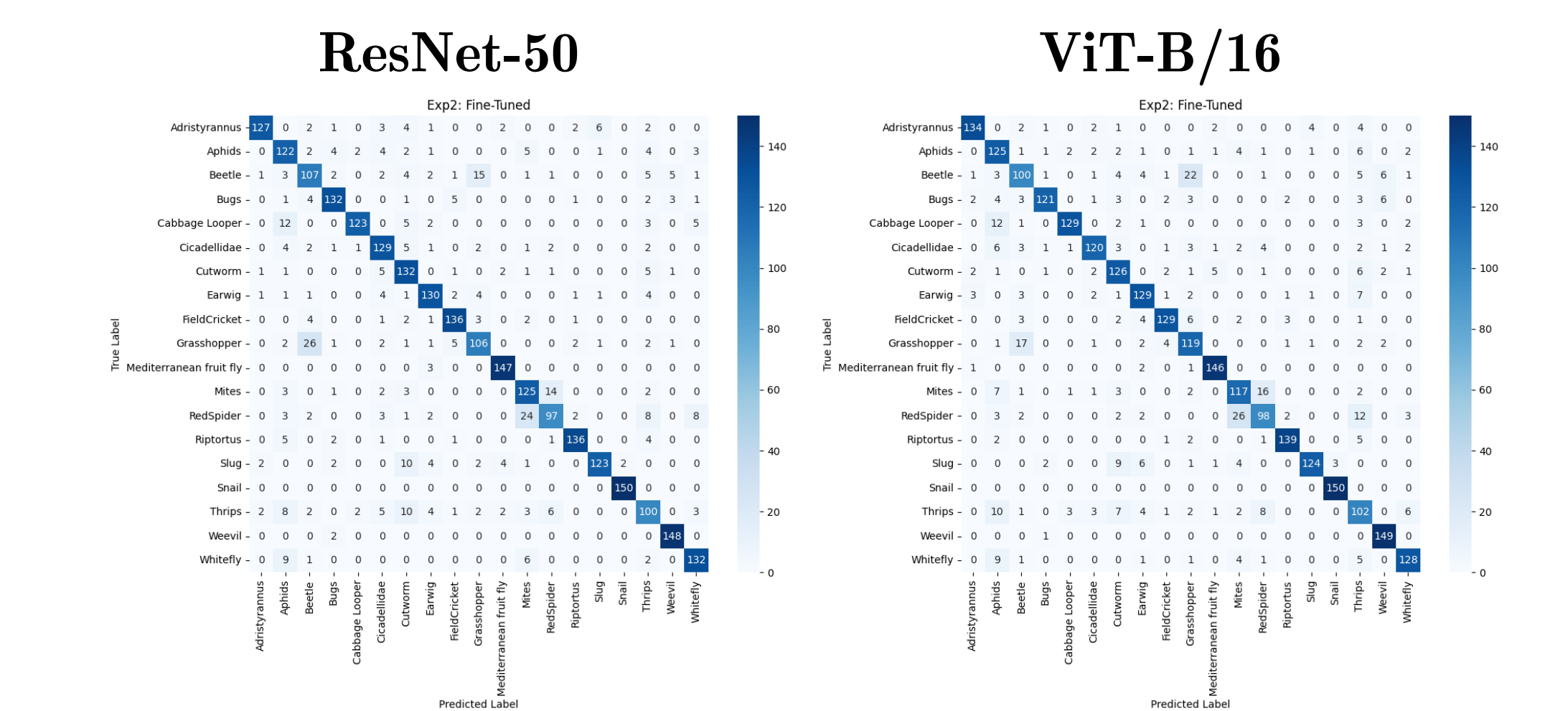


Accuracy: 0.0477
Macro F1 Score: 0.0348

Accuracy: 0.0589
Macro F1 Score: 0.0546

Consistent with the first dataset, both models performed poorly on Pest Dataset V2, effectively making random predictions, as seen in the confusion matrices. ResNet-50 predominantly predicted Riptortus, while ViT-B/16 produced more scattered predictions, both showing common misclassifications (e.g. Weevil/Snails, Weevil/Cicadellidae). Overall, ViT-B/16 performed slightly better, suggesting improved performance on larger datasets.

Exp 2: Fine-Tuned (Pest Dataset V2)



Accuracy: 0.8411
Macro F1 Score: 0.8403
Early Stopping at Epoch 7

Accuracy: 0.8428
Macro F1 Score: 0.8428
Early Stopping at Epoch 6

After fine-tuning, both models showed a significant improvement in performance compared to experiment 1 and greater accuracy compared to the smaller dataset, as expected. Both models now consistently predict the correct pest indicated by the dark diagonals on the confusion matrices, with no clear commonly mistaken pests. Performance was similar across both models, with no significant difference, however, ViT-B/16 learned faster but took longer.

Conclusion

Pretrained models alone (with no additional training) do not generalize well to real-world pest datasets. However, with training, both model types (ResNet-50 and ViT-B/16) perform relatively well, with the vision transformer overall generalizing slightly better, indicating the importance of needing to adapt models to domain-specific data.